



PROJECT DOCUMENTATION

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM

USING AGENTFORCE

Salesforce | Auto-Launched Flow | Agentforce

TEAM DETAILS

TEAM ID	SWTID-2026-7679
TEAM SIZE	4
TEAM LEADER	Priyadharshini J NMID: CB0E45A5610E3B7A80D4E9F2F74B93FA
TEAM MEMBER	Susmitha R NMID: 42C35A146C4BF81B09B4C6771D29B681
TEAM MEMBER	Srimathi R NMID: 137C65A061B7F819292474B958E38893
TEAM MEMBER	Pushpa A NMID: 48F149E0269F3E28F7D4003F6F884D04

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1. Introduction

Project Overview

Support teams handle a steady stream of tickets, and deciding which ones need attention first is usually done by hand. This project replaces that manual step with a Salesforce solution. A custom object called **Support Ticket Intelligence** stores each ticket, an **Auto-Launched Flow** reads the latest ticket of a customer Account and classifies it as High, Medium or Low priority, and an **Agentforce** subagent lets a support user trigger the whole process by simply typing an Account Name.

When a ticket is classified as High, the flow also creates an urgent handling task and routes the ticket to a senior support agent, so critical issues stop waiting in the queue.

Purpose

- Classify support tickets as High, Medium or Low priority without manual review.
- Fetch the most recent ticket that belongs to a given customer Account.
- Create an Urgent Ticket Handling task automatically for High-priority tickets.
- Assign the right support level (for example, a senior agent for urgent cases).
- Give support users a conversational way to run the analysis through Agentforce.
- Cut response time and the repetitive effort spent on triage.

2. Ideation Phase

Problem Statements

The table below records the problems that motivated the project, written from the point of view of each person affected.

#	I am	I am trying to	But	Because	Which makes me feel
PS-1	A support agent	Focus on critical tickets first	Tickets are not always prioritized automatically	Prioritization is largely manual	Frustrated
PS-2	A customer	Get urgent issues resolved quickly	Critical issues can wait behind minor ones	Teams handle very large ticket volumes	Concerned
PS-3	A support manager	Assign tickets to the right support level	Assignment needs manual decisions	Urgency is not analyzed consistently	Overwhelmed
PS-4	A support agent	Spot urgent issues from descriptions	Important keywords get missed	Descriptions carry different urgency signals	Uncertain
PS-5	A support manager	Reduce SLA-related risk	Older unresolved tickets go unnoticed	SLA risk needs an extra check	Concerned

Summary of needs. High ticket volume, manual prioritization, wrong or slow assignment, delayed resolution and a poor customer experience all point to one need: a smart, automated system that classifies priority, routes tickets and shortens response time.

Empathy Map

Persona: Support Agent. The map captures what the agent experiences while working through the queue.

THINKS & FEELS	Wants critical issues identified quickly so urgent customer problems get attention.
HEARS	Customer issue descriptions and support requests arriving all day.
SEES	Ticket status, issue type, description, priority level, assignment and SLA risk.
SAYS & DOES	Reviews tickets, handles assigned cases and focuses on urgent support issues.
PAINS	Manual prioritization, missed urgency signals and delays in handling critical tickets.
GAINS	Automatic priority classification, clearer assignment, urgent-task creation and less manual effort.

Brainstorming and Idea Prioritization

Four approaches were considered: manual ticket registers, spreadsheet-based tracking, a standalone support portal, and Salesforce CRM combined with Agentforce. Ideas were then placed on an impact-versus-feasibility grid.

	LOW FEASIBILITY	HIGH FEASIBILITY
HIGH IMPACT	Use AI to classify ticket priority; customer sentiment analysis for urgency	Automatically assign tickets to the right agent; real-time notifications for urgent tickets; SLA breach alerts; dashboards for ticket trends
LOW IMPACT	Mobile access for support agents	Self-service FAQs; automatic email updates; knowledge article suggestions; automation to update status; balance workload across agents

Top prioritized ideas: AI-based priority classification, automatic ticket assignment, real-time notifications, dashboards and reporting, and SLA breach alerts.

Final Selection

Salesforce CRM with Agentforce was chosen because it offers:

- Automation through Salesforce Flow
- AI-assisted ticket analysis through Agentforce
- Automatic priority classification
- Automatic urgent-task creation
- Less manual support effort

3. Requirement Analysis

Customer Journey Map

A ticket moves through eight stages, from the moment a customer reports an issue to the feedback collected after resolution.

Stage	Customer action	System / AI action	Outcome
1. Raise ticket	Submits a ticket with issue details	Creates the ticket record in Salesforce	Ticket created
2. Ticket received	Waits for confirmation	Stores the ticket and triggers automation	Ticket logged
3. AI analysis	No direct action	Analyzes the description and classifies priority	Priority identified
4. Auto assignment	No direct action	Assigns the ticket to the right agent	Ticket assigned
5. Agent works	May add information	Agent updates status, adds notes, resolves	Issue resolved
6. Updates	Receives status updates	Sends automated notifications	Customer informed
7. Resolved	Confirms resolution	Closes the ticket	Ticket closed
8. Feedback	Shares feedback and rating	Captures feedback for improvement	Insights gathered

Solution Requirements

Functional requirements

FR No	Requirement
FR-1	Support ticket creation and management
FR-2	Account and Contact association
FR-3	Ticket description analysis
FR-4	Automatic priority classification
FR-5	High-priority task creation
FR-6	Support agent assignment
FR-7	SLA breach risk checking
FR-8	Agentforce conversational analysis

Non-functional requirements

Attribute	Description
Usability	Clear Agentforce interaction and easy-to-read priority messages
Reliability	Consistent classification based on the configured conditions
Performance	Quick retrieval and analysis of the latest support ticket
Security	Controlled Salesforce access to customer and ticket information
Maintainability	Flow and Agentforce configuration can be updated without redevelopment
Scalability	Handles growing ticket volumes through Salesforce automation

Data Flow

From the request to the final response, data moves through the following stages:

Customer / support request	Account	Latest support ticket	Ticket description
Analyze description	Priority level	High / Medium / Low decision	Task + agent assignment

Through Agentforce the same path starts with a conversation:

User gives Account Name	Agentforce subagent	Auto-Launched Flow
Ticket analysis	Priority and assignment output	Response to user

Technology Stack

Layer	Technology
Platform	Salesforce (Developer Edition and Trailhead Playground orgs)
Data	Support_Ticket_Intelligence__c, with Account, Contact, User and Task records
Logic	Auto-Launched Flow
AI	Agentforce (Default) with a custom subagent
Agent configuration	Support Ticket Priority Analysis subagent
Automation	Flow Decision, Assignment and Create Records elements
Output	Agentforce conversation response

4. Project Design

Problem-Solution Fit

Problem	Solution	Benefit
Manual ticket prioritization	Flow analyzes the description and sets priority	Faster identification of urgent cases
Urgent issues may be missed	High-priority keyword decision	Critical cases get attention
Manual task creation	Flow creates a Task for High priority	Less manual effort
Manual support assignment	Flow sets the assigned support level	More consistent handling
SLA risk can be overlooked	Optional SLA risk decision	Better risk awareness
Need for conversational access	Agentforce subagent interaction	AI-assisted usage

Proposed Solution

Parameter	Description
Problem	Manual prioritization and assignment delays the handling of critical customer support issues.
Solution	A Salesforce-based ticket priority prediction and automated assignment system.
Innovation	Agentforce AI combined with an Auto-Launched Flow to analyze descriptions, classify priority, assign a support level and trigger urgent-task automation.
Impact	Critical tickets are found sooner, manual effort drops, response time improves and the team is more productive.
Business model	Customer support operations involving customers, agents, managers and AI-assisted ticket handling.
Scalability	Scales with ticket volume through Salesforce automation, with room for more assignment rules, SLA monitoring and priority indicators.

Solution Architecture

The solution is organized in four layers. A request enters through Agentforce, the Auto-Launched Flow does the analysis, and Salesforce records hold the data and the results.

Users	Support agents Support managers Customers (via tickets)
AI layer	Agentforce (Default) Support Ticket Priority Analysis subagent Flow-based agent action
Automation layer	Get Account / Get Ticket Analyze Description decision Create Task, assign agent, set message
Data layer	Account and Contact Support_Ticket_Intelligence__c Task and User

5. Project Planning and Scheduling

Planning Approach

- Agile methodology with sprint-based execution
- Epics are broken into user stories and sized in story points
- Velocity is calculated from completed story points and used for estimation

Product Backlog and Sprint Schedule

Sprint	Epic	USN	User story / task	Pts	Priority	Owner
1	Developer setup	USN-1	As a system administrator, I want to configure the Salesforce environment for Support Ticket Intelligence so the automation can be developed.	3	High	Priyadharshini J
2	Data modeling	USN-2	As an admin, I want to create the Support Ticket Intelligence object with the required fields and relationships so ticket data is stored systematically.	5	High	Susmitha R
2	Data modeling	USN-3	As a support user, I want tickets associated with Accounts and Contacts so the latest ticket of a customer can be identified.	3	High	Srimathi R
3	Automation	USN-4	As a support agent, I want an Auto-Launched Flow to retrieve the latest ticket and analyze its description so priority can be set automatically.	5	High	Pushpa A
3	Automation	USN-5	As a support agent, I want tickets classified as High, Medium or Low from configured urgency conditions so critical issues are handled first.	5	High	Priyadharshini J
4	Agentforce and automation	USN-6	As an AI agent, I want to analyze ticket details and trigger the backend Flow so manual prioritization is reduced.	8	High	Susmitha R
4	Task and assignment	USN-7	As a support manager, I want High-priority tickets to create an urgent task and be assigned to the right support level.	5	High	Srimathi R
5	SLA management	USN-8	As a support manager, I want the system to check SLA breach risk for older unresolved tickets.	5	Medium	Pushpa A
6	Conversational support	USN-9	As a support user, I want to give an Account Name to Agentforce and receive the ticket ID, priority, assigned agent and action message.	5	Medium	Priyadharshini J

Story-point total: 44 across six sprints (3 + 8 + 10 + 13 + 5 + 5).

Sprint Tracker and Velocity

Sprint	Story points	Stories	Points completed
Sprint-1	3	USN-1	3
Sprint-2	8	USN-2, USN-3	8
Sprint-3	10	USN-4, USN-5	10
Sprint-4	13	USN-6, USN-7	13
Sprint-5	5	USN-8	5
Sprint-6	5	USN-9	5

Average velocity = total story points / number of sprints = 44 / 6 = about 7.3 points per sprint.

6. Project Development

The topics covered while building the solution were data modeling, fields and relationships, the Lightning App Builder, Salesforce Flow and Agentforce configuration.

Milestone 1: Salesforce Environment

Activity 1: Create the Trailhead Playground

A hands-on org was created from the Hands-On Orgs page in Trailhead using Create Playground. The playground **Empathetic Badger Playground** (created 10/1/2026, type Trailhead Playground) is listed under Connected Orgs.

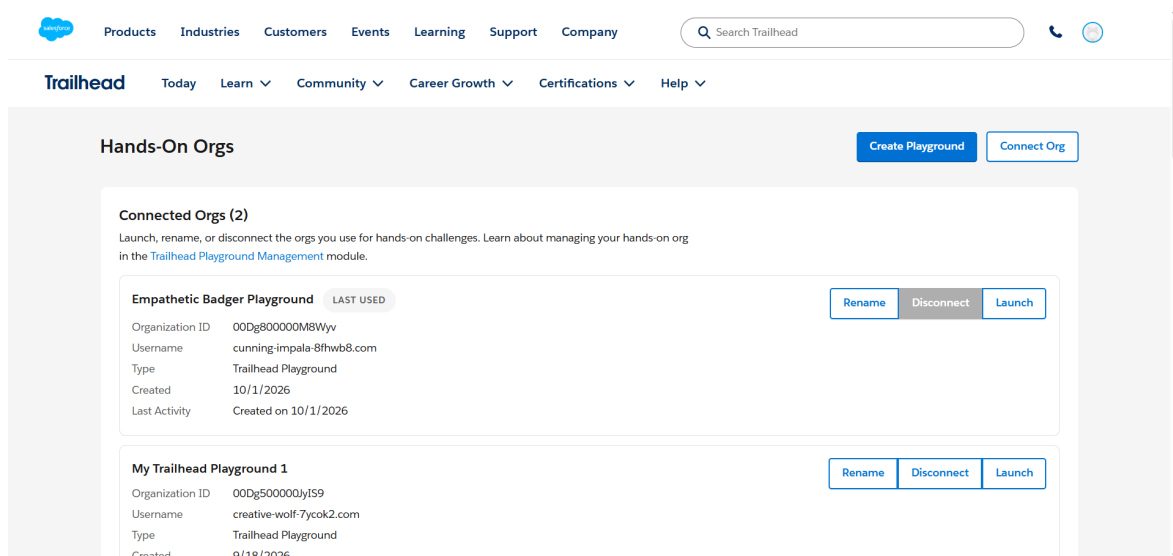


Figure 1. Trailhead Hands-On Orgs page listing the connected orgs, with Empathetic Badger Playground as the last used.

Activity 2: Launch the org

Selecting **Launch** opens the org and signs in as the team leader, Priyadharshini J. The Playground Starter app opens on its Welcome page.

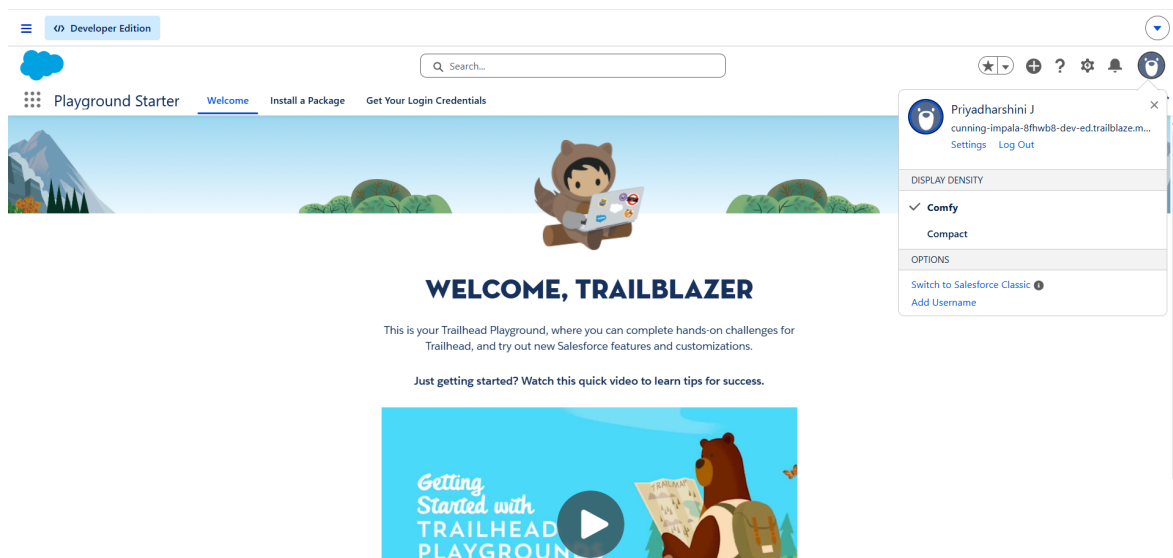


Figure 2. Playground Starter welcome page with the signed-in user menu open.

Activity 3: Open Setup

All configuration in the next milestones is done from **Setup**. The org is a Developer Edition org, used for the Agentforce work.

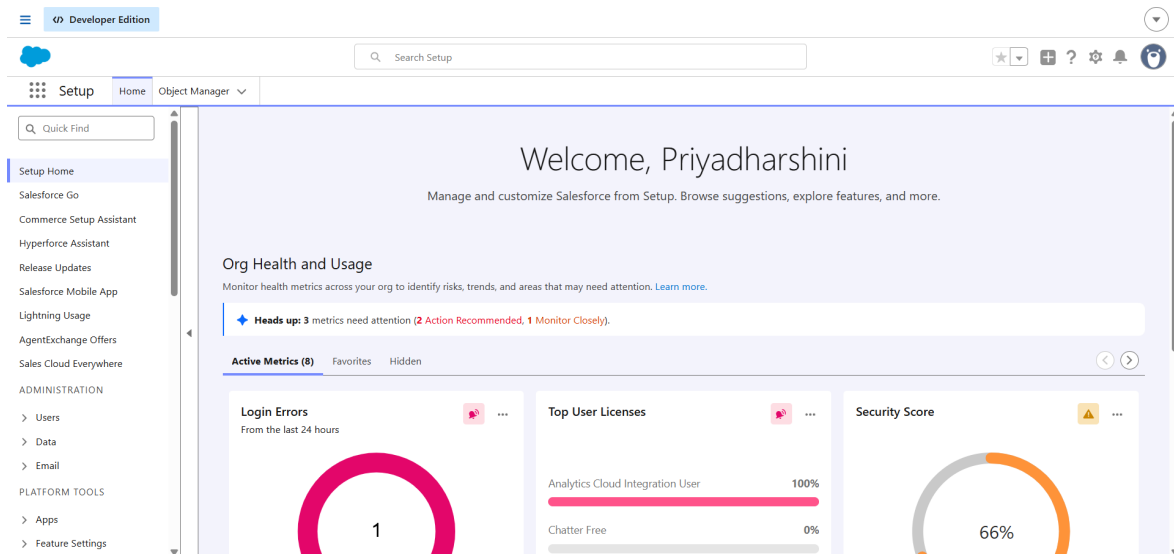


Figure 3. Setup Home of the Developer Edition org.

Milestone 2: Data Model

Custom object

Property	Value
Object label	Support Ticket Intelligence
API name	Support_Ticket_Intelligence__c
Record name	Ticket Number (Auto Number, format TKT-{0000})

Fields

Field label	API name	Data type	Description
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related Account
Contact	Contact__c	Lookup (Contact)	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken

Sample records

A test ticket, **TKT-0001**, was created so the flow and the agent had data to work on. Its customer is Edge Communications, the contact is Rose Gonzalez, the issue type is Technical, the priority level is High and the status is In Progress.

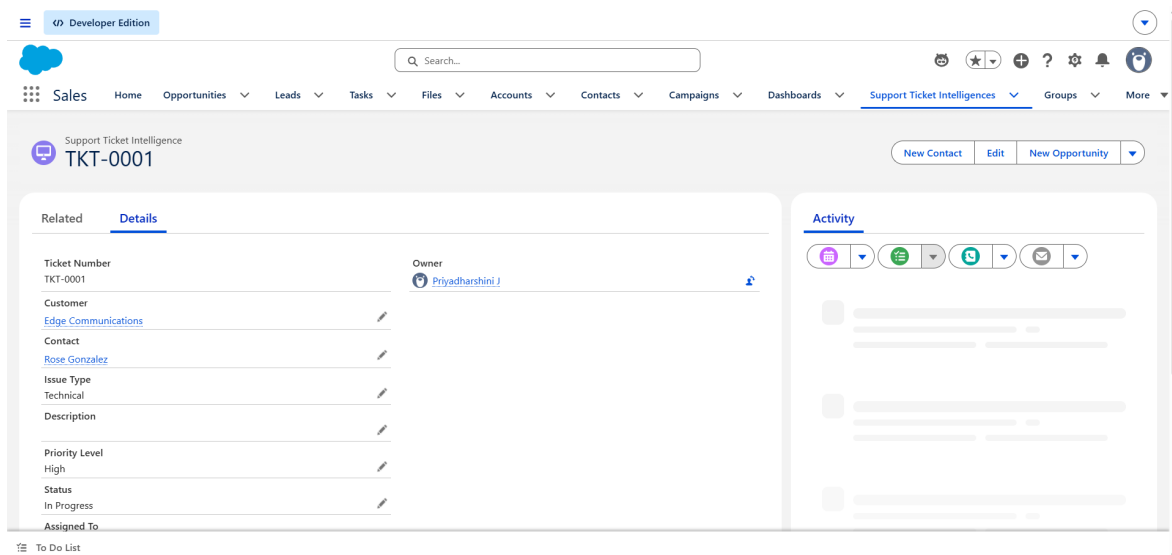


Figure 4. Detail page of ticket TKT-0001: customer, contact, issue type, priority level, status and owner.

Milestone 3: Auto-Launched Flow

The flow takes an Account Name, finds the Account, fetches its latest ticket, checks the description for urgency keywords and returns the priority, the assigned agent and a message. It is an **Autolaunched Flow** with no trigger so that Agentforce can call it as an action. The flow is saved as **Support Ticket Intelligence** and appears in the Flows list as an unmanaged, active flow.

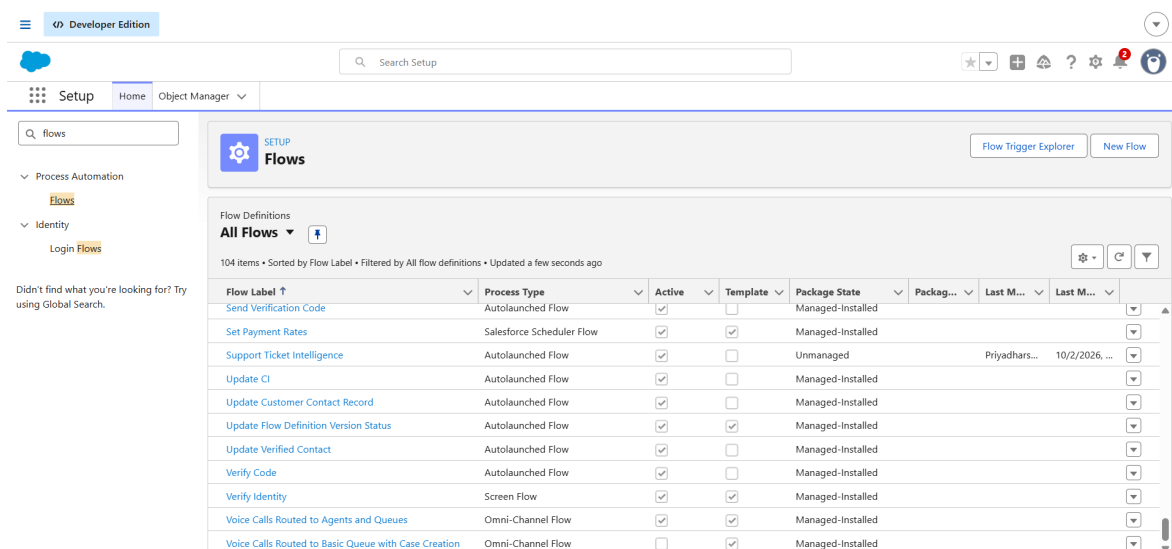


Figure 5. Flows list in Setup showing the Support Ticket Intelligence autolaunched flow (Autolaunched Flow, Active, Unmanaged).

Flow variables

Label	API name	Type	Direction
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

In addition, flow constants (such as varUrgent, varIssue, varAgent, varSeniorAgent, varMediumAgent, varMediumMessage and varFinalMessage) hold the keywords, agent names and messages used by the flow, so they can be changed in one place.

Flow logic

Step	Element	Configuration
1	Get Records: Get Account	Object Account, Name equals {!varAccountName}, sorted by CreatedDate descending, first record only
2	Assignment: Store Account Id	varAccountId = Account Id from step 1
3	Get Records: Get Ticket	Object Support_Ticket_Intelligence__c, Customer__c equals {!varAccountId}, newest first, first record only
4	Assignment: Store Ticket Id	varTicketId = Id of the ticket found in step 3
5	Decision: Analyze Description	High: description contains "urgent", "not working" or "failure". Medium: contains "issue", "slow" or "delay". Default: Low
6	Assignment: High / Medium / Low Path	varPriorityLevel is set to High, Medium or Low
7	Create Records: Create Task (High only)	Subject "Urgent Ticket Handling", WhatId = varTicketId, Priority High, Status Not Started
8	Assignment: Set Assigned Agent	varAssignedTo = "Senior Support Agent" for High priority
9	Decision (optional): SLA risk	Flags tickets whose created date is older than two days
10	Assignment: final message	High: "High priority ticket detected. Assigned to senior agent." Medium: "Ticket marked as medium priority. Will be handled shortly." Low: "Tickets are low priority and queued for processing."

Flow Builder canvas

The top of the flow retrieves the Account and the ticket and stores their Ids.

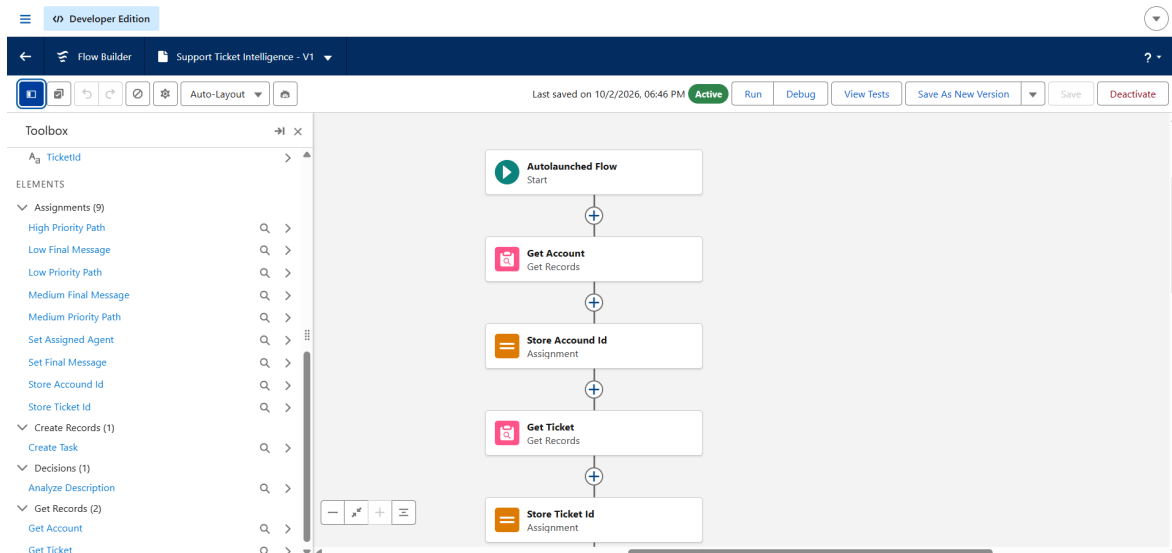


Figure 6. Flow Builder: Autolaunched Flow start, Get Account, Store Account Id, Get Ticket and Store Ticket Id.

After the Analyze Description decision the flow splits into three paths. The High path creates the task and sets the senior agent, while the Medium and Low paths only set their final messages before ending.

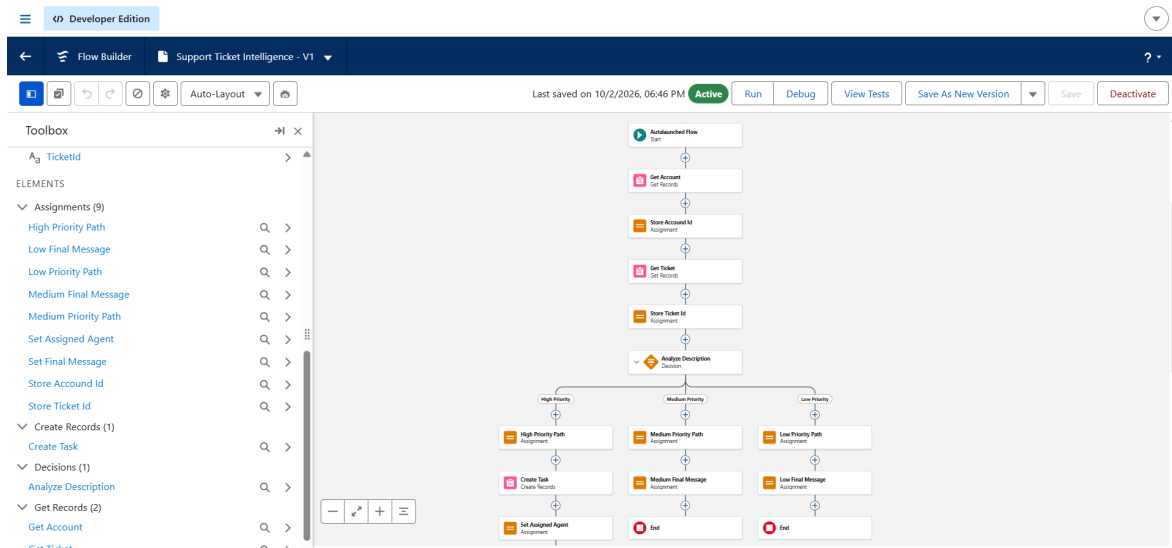


Figure 7. Full flow (version 1, Active) with the High, Medium and Low priority branches, Create Task and Set Assigned Agent on the High path.

Milestone 4: Making the Object Available in the Sales App

To let users open tickets from the Sales app, the object was added to its navigation bar: **Edit navigation items > Add More Items > Support Ticket Intelligences > Save**. The new tab now appears in the Sales app, and its list view shows the ticket records.

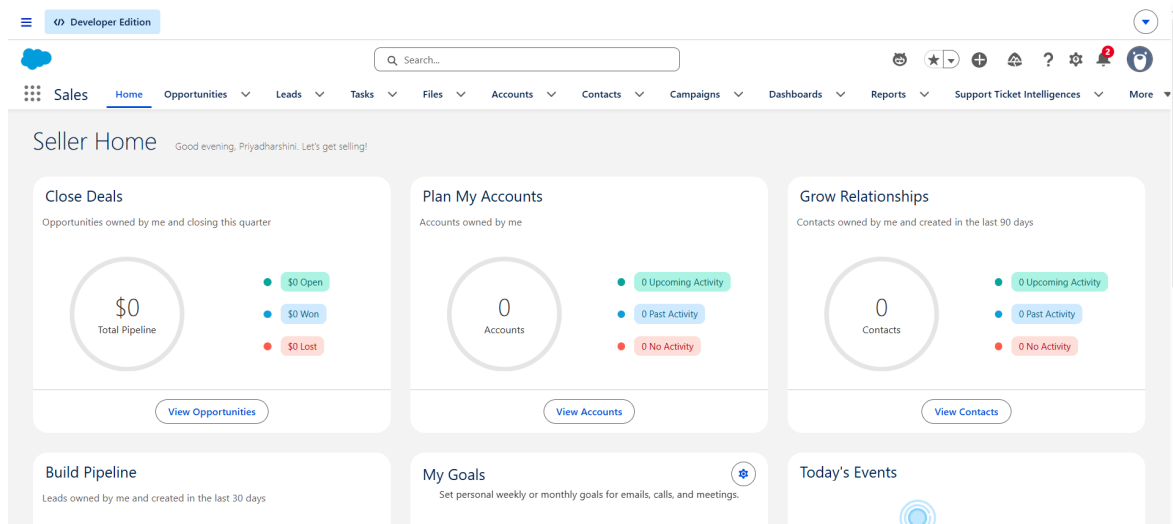


Figure 8. The Support Ticket Intelligences tab in the Sales app navigation bar on Seller Home.

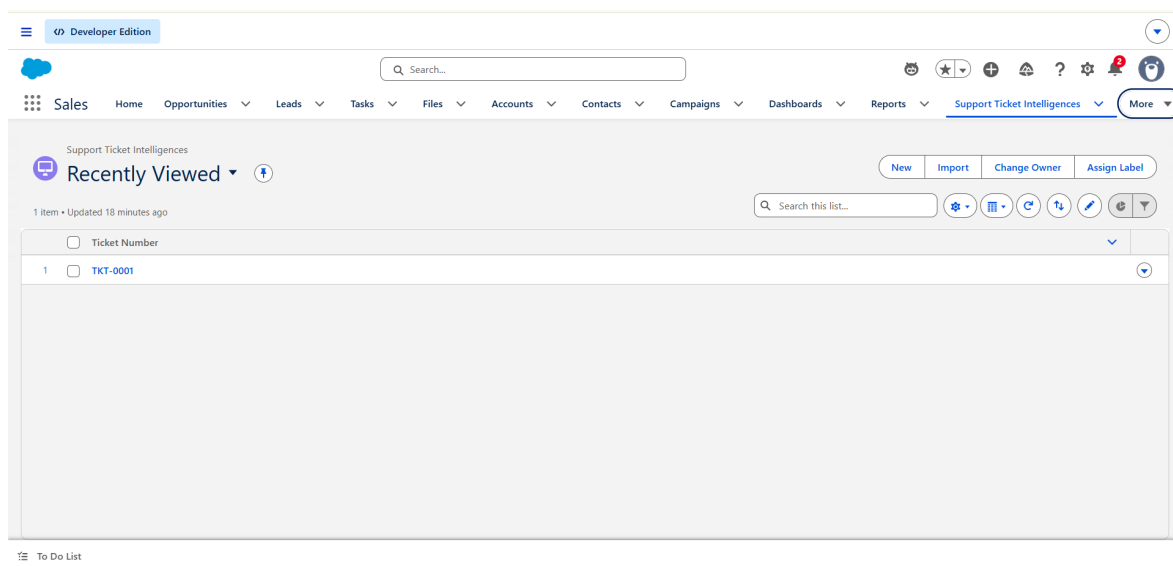


Figure 9. Support Ticket Intelligences list view (Recently Viewed) showing TKT-0001.

Milestone 5: Agentforce Configuration

Turn on Agentforce

1. Open the gear menu and choose **Salesforce Go**.
2. Search for "default", select **Agentforce (Default)** and open it.
3. Choose **Get Started > Turn On > Confirm**.
4. In Setup search for **Agents**, make sure Agentforce is enabled and open **Agentforce (Default)**.

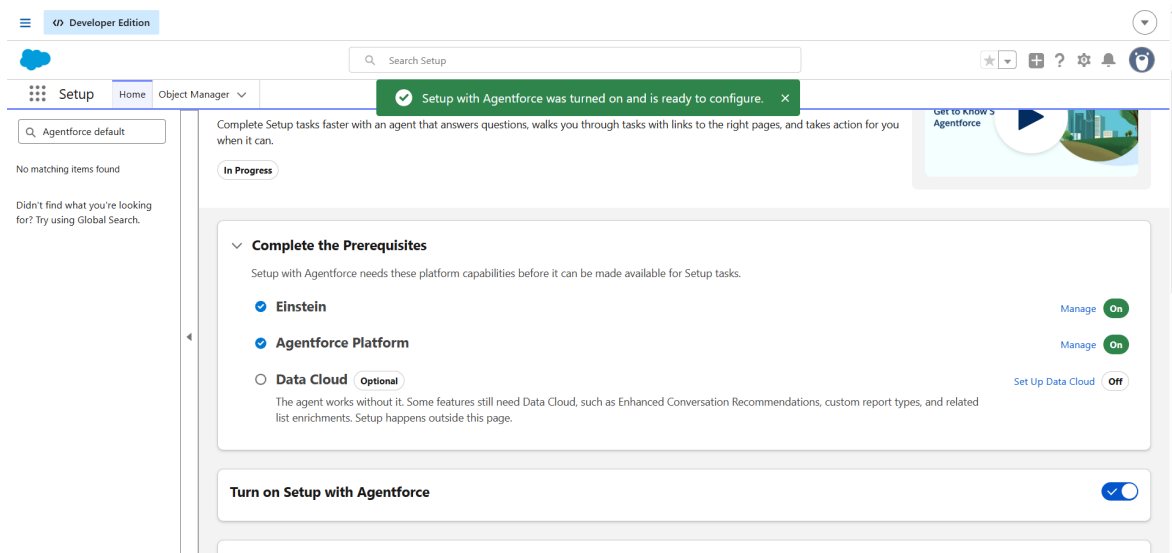


Figure 10. Setup with Agentforce prerequisites (Einstein and Agentforce Platform turned on) and Setup with Agentforce turned on and ready to configure.

Open the builder and add a subagent

1. Select the default agent and click **Open in Builder**.
2. In the Subagents panel click **New > New Subagent**, then **Next**.
3. Fill in the name, description, scope and instructions given below, then click **Next**.
4. Choose the flow action and click **Finish**.

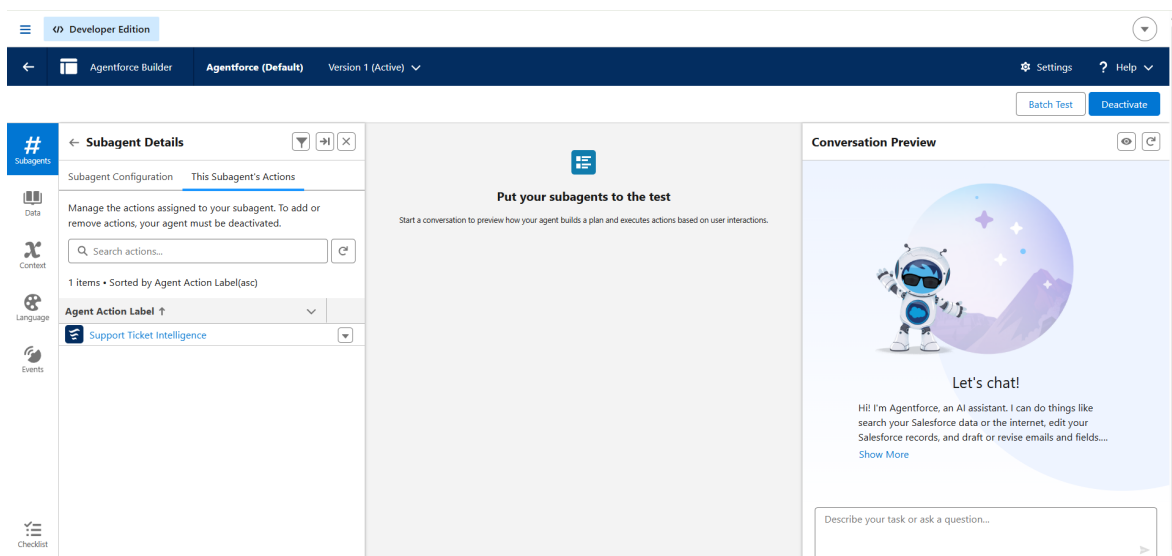


Figure 11. Agentforce Builder for Agentforce (Default), Version 1 (Active), with the subagent actions listing the Support Ticket Intelligence action and the conversation preview.

Subagent configuration

Property	Value
Name	Support Ticket Priority Analysis
API name	Support_Ticket_Priority_Analysis
Classification description	Analyzes customer support tickets to determine priority from the issue description and urgency. Handles requests about ticket status, issue severity and support prioritization.
Scope	Only analyze support ticket descriptions, determine the priority level (High, Medium, Low) and trigger backend automation for task assignment. Do not handle unrelated requests such as billing, subscription management or account updates.

Instructions given to the subagent

1. Ask the user for the Account Name if it has not been given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Decide the priority from keywords: High for "urgent", "not working" or "failure"; Medium for "issue", "slow" or "delay"; otherwise Low.
5. Set the priority level accordingly.
6. For High priority, trigger the Auto-Launched Flow and make sure an immediate-handling task is created.
7. Assign the ticket to the appropriate support level (a senior agent for High priority).
8. Reply with a clear message: High means urgent handling has started, Medium means it will be handled shortly, Low means the ticket is queued.
9. Never update records directly or send emails; use the Flow for every action.

Agent action

The action references the flow (Reference Action Type: Flow) with the label **Support Ticket Intelligence**. Its description reads: Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases. Loading text: Analyzing ticket details and determining priority...

Variable	Role	Purpose	Settings
varAccountName	Input	Customer account name used to fetch the latest ticket	Require input and collect from user
varAccountId	Output	Id of the Account used for the lookup	Show in conversation, text
varTicketId	Output	Id of the analyzed ticket	Show in conversation, text
varPriorityLevel	Output	Calculated priority: High, Medium or Low	Show in conversation, text
varAssignedTo	Output	Agent or team assigned to the ticket	Show in conversation, text
varActionMessage	Output	Final message describing priority and actions taken	Show in conversation, text

7. Functional and Performance Testing

The agent was tested from the Conversation Preview inside Agentforce Builder. The preview also shows the reasoning trace, including which subagent was selected for each message.

Test	Input	Expected result	Evidence
TC-01	Greeting: "hi"	Agent introduces itself and offers help; a subagent is selected	Figure 12
TC-02	"what is the support ticket priority?" then Account Name	Agent asks for the Account Name, selects the Support Ticket Priority Analysis subagent, runs the flow action and replies with the priority message	Figure 13
TC-03	Ticket text containing "urgent", "not working" or "failure"	Priority High, task created, senior agent assigned	Flow logic, Section 6
TC-04	Ticket text containing "issue", "slow" or "delay"	Priority Medium, handled shortly message	Flow logic, Section 6
TC-05	Ticket text with none of the keywords	Priority Low, queued message	Flow logic, Section 6

Agent greeting

The agent greeted the user ("Hello! How can I assist you today?"). The reasoning trace shows the Single Record Summary subagent was selected for the greeting, which is the default agent behaviour for a message with no ticket request.

Figure 12. Conversation Preview: session start, greeting and subagent selection.

Priority analysis through the custom subagent

For the question "what is the support ticket priority?", the reasoning trace shows the **Support Ticket Priority Analysis** subagent was selected. Its description, 9 instructions and 1 action are visible in the trace. The agent asked for the required AccountName input, the value "Support Ticket Intelligence" was submitted, and the flow action returned the message "Tickets are low priority and queued for processing." with the priority Low.

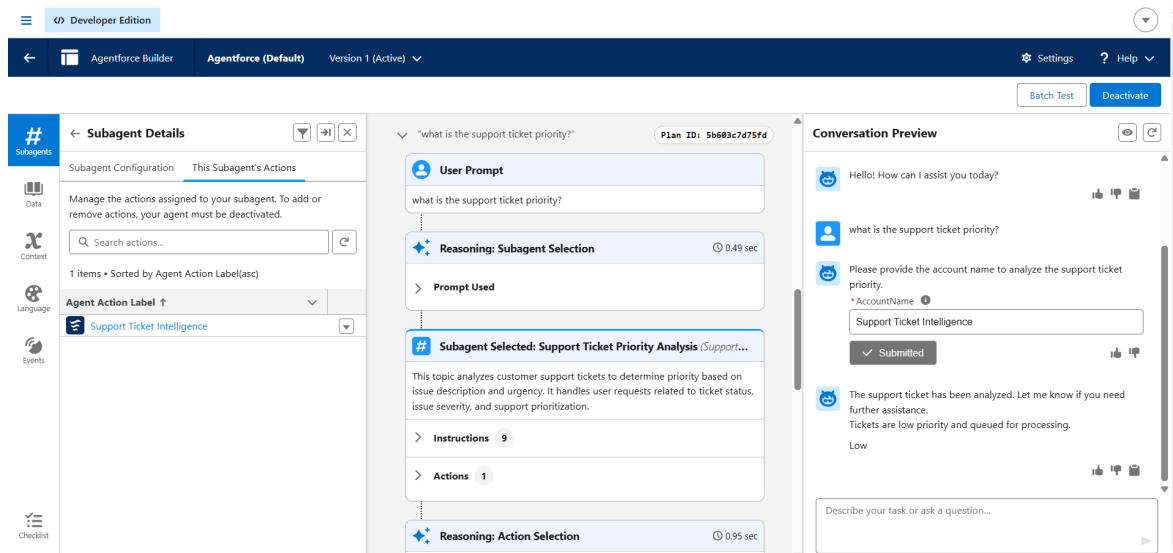


Figure 13. Agentforce response for the priority question, with the reasoning trace on the left.

Observation from testing

The flow classifies priority from the ticket *description* text, not from the Priority Level field. In the test above, the Account Name entered was not the customer of TKT-0001 (Edge Communications), and TKT-0001 has an empty Description. With no matching keyword, the flow correctly falls to its Low default. Re-running the test with the Account Name **Edge Communications** and keyword-bearing descriptions (for example "urgent" or "slow") is the way to exercise the High and Medium paths, TC-03 and TC-04.

8. Advantages and Disadvantages

Advantages

- Automatic prioritization of tickets as High, Medium or Low from the description.
- Faster handling of critical issues because agents see urgent tickets first.
- Less manual effort in analysis, priority setting, task creation and assignment.
- Urgent task creation for every High-priority ticket.
- Better assignment, with High tickets going to a senior agent.
- Conversational access through Agentforce.
- SLA awareness through the optional check on older tickets.
- Scalable with ticket volume through Salesforce automation.
- Higher team productivity as repetitive triage is removed.

Disadvantages

- Keyword-based logic that depends on words such as urgent, not working, failure, issue, slow and delay.
- Limited context understanding; some urgency signals can be missed.
- Depends on accurate data for Account, Contact and ticket details.
- Flow and Agentforce maintenance is needed when requirements change.
- Narrow agent scope; billing and subscription requests are out of scope.
- Limited SLA monitoring; only an optional risk check, not full escalation.
- Assignment rules need expansion for more teams.
- Limited analytics; trend and resolution reporting is future scope.

9. Conclusion

The project shows how Salesforce Flow and Agentforce can work together to take the guesswork and repetition out of ticket triage. Given an Account Name, the solution finds the latest ticket, classifies its priority from the description, creates an urgent task for High-priority cases, assigns a support level and replies in plain language. The result is faster handling of critical issues, less manual work and a more productive support team.

10. Future Scope

- Add more priority indicators and business rules.
- Return additional ticket information from the Agentforce action.
- Build fuller SLA monitoring and escalation.
- Support more support teams and assignment rules.
- Add analytics for priority trends and resolution performance.
- Add more flow variables and map them in the agent action for richer answers.

11. Appendix

Ref	Item	Description
A	Custom object	Support Ticket Intelligence (Support_Ticket_Intelligence__c)
B	Key Salesforce records	Account, Contact, User, Task
C	Flow	Support Ticket Intelligence, Auto-Launched Flow
D	Agentforce subagent	Support Ticket Priority Analysis
E	Agentforce action	Flow-based Support Ticket Intelligence action